

**E2 Small angle
approximations A level
only**

Name: _____

Class: _____

Date: _____

Time: **15 minutes**

Marks: **12 marks**

Comments:

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Q1.

Show that, for small values of x , the graph of $y = 5 + 4\sin\frac{x}{2} + 12\tan\frac{x}{3}$ can be approximated by a straight line.

(Total 3 marks)

Q2.

When θ is small, find an approximation for $\cos 3\theta + \theta \sin 2\theta$, giving your answer in the form $a + b\theta^2$

(Total 3 marks)

Q3.

$$f(x) = \sin x$$

Using differentiation from first principles find the exact value of $f'\left(\frac{\pi}{6}\right)$
Fully justify your answer.

(Total 6 marks)



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